homework$_1$

June 12, 2026

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(1) If R is a ring, show that $M_n(R)$ is a ring under the usual operations of matrix addition and multiplication.

(2) Let $(G, +)$ be an Abelian group. An endomorphism of G is a group homomorphism $\varphi : G \to G$. Let $\mathrm{End}(G)$ denote the set of all endomorphisms on G. Given $\varphi, \psi \in \mathrm{End}(G)$, define

$$(\varphi + \psi)(g) := \varphi(g) + \psi(g), \quad \text{and} \quad (\varphi \times \psi)(g) := \varphi(\psi(g)).$$

(a) Show that $+$ and $\times$ are well-defined binary operations on $\mathrm{End}(G)$.

(b) Show that $(\mathrm{End}(G), +, \times)$ is a ring.

(3) Let R be a ring and $a, b \in R$. Show that

(a) $0 \cdot a = a \cdot 0 = 0$.

(b) For any $n \in \mathbb{Z}$, $(na)b = n(ab) = a(nb)$. [Hint: Use Induction.]

(c) If R is unital, then the multiplicative identity is unique. Moreover, $(-1) \cdot a = (-a)$ for all $a \in R$.

(4) Let R be a commutative ring and let S and T be two subsets of R. Define

$$ST := \bigcup_{n=1}^{\infty} \left\{ \sum_{i=1}^{n} s_i t_i : s_i \in S, t_i \in T \right\}.$$

(In other words, elements in ST are finite sums of the form $\sum_{i=1}^{n} s_i t_i$, where $s_i \in S$ and $t_i \in T$, while the number n may vary.)

(a) If S and T are both subrings of R, prove that ST is a subring.

(b) If S and T are both ideals, then prove that ST is an ideal.

(5) Fix $m, n \in \mathbb{Z}$ and let $d := \gcd(m, n)$. If I denoted the ideal generated by $\{m, n\}$, then show that $I = (d)$, the principal ideal generated by d (See Definition 5.3).

Extra Problems

You need not submit these problems; they are here only for practice.

Definition: A function $f : \mathbb{R} \to \mathbb{R}$ is said to have **compact support** if there exists $a, b \in \mathbb{R}$ such that $f(x) = 0$ for all $x \notin [a, b]$.

Define

$$C_c(\mathbb{R}) := \{f : \mathbb{R} \to \mathbb{C} : f \text{ is continuous and has compact support}\}.$$

Equip $C_c(\mathbb{R})$ with the pointwise operations as in Part (v) of Example 1.2 in the notes. You need not check that $C_c(\mathbb{R})$ is a ring (this is easy to do).

1. Prove that $C_c(\mathbb{R})$ is not unital.

2. Let R be a ring, S be a subring of R and let I be an ideal of R.
   (a) Prove that $S + I := \{s + a : s \in S, a \in I\}$ is a subring of R.
   (b) Prove that $I \lhd S + I$.
   (c) Prove that $S \cap I \lhd S$.
   (d) Prove that
   $$\frac{S+I}{I} \cong \frac{S}{S \cap I}$$
   [Hint: Define $\varphi : S \to (S+I)/I$ by $s \mapsto s + I$.]

3. Let $D \in \mathbb{Q}$ be a rational number that is not a perfect square. Define
   $$R := \{a + b\sqrt{D} : a, b \in \mathbb{Q}\}.$$
   (a) Show that R is a subring of $\mathbb{C}$.
   (b) If $(a + b\sqrt{D}) \neq 0$, then show that $a^2 - b^2 D \neq 0$.
   (c) Use Part (b) to prove that R is a field. This field is denoted $\mathbb{Q}(\sqrt{D})$, and is called a **quadratic field**.

4. For a unital ring R, define the **center** of R to be
   $$\mathcal{Z}(R) := \{a \in R : ab = ba \text{ for all } b \in R\}.$$
   Show that $\mathcal{Z}(R)$ is a subring of R.

Given the handwritten notes in the image, —

1. Given: R is a ring. Show: $M_n(\mathbb{C}R)$ is a ring under C^+, matrix multiplication.

   Proof. (iii) Fix $n \in \mathbb{N}$. Show: $M_n(\mathbb{C}R)$ is a ring under addition and multiplication of real numbers is $C = AB$ such that
   $$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj} \quad \text{(Check)}$$

   Associativity of multiplication:
   $$A \cdot (B \cdot C) = A \cdot D \quad \text{where} \quad d_{ij} = \sum_{k=1}^{n} b_{ik} c_{kj}$$
   $$\Rightarrow A \cdot (B \cdot C) = \left(\sum_{\ell=1}^{n} a_{i\ell} d_{\ell j} \right)_{n \times n}$$
   $$= \left(\sum_{\ell=1}^{n} a_{i\ell} \left(\sum_{k=1}^{n} b_{\ell k} c_{kj} \right) \right)_{n \times n}$$
   $$= \left(\sum_{k=1}^{n} \left(\sum_{\ell=1}^{n} a_{i\ell} b_{\ell k} \right) c_{kj} \right)_{n \times n}$$

$$= (A \cdot B) \cdot C$$

Distributive Property:

$$AB + AC = \left(\sum_{k=1}^{n} a_{ik} b_{kj} + \sum_{k=1}^{n} a_{ik} c_{kj} \right)_{n \times n}$$

$$= \left(\sum_{k=1}^{n} a_{ik}(b_{kj} + c_{kj}) \right)_{n \times n}$$

$$= A(B + C)$$

Similarly, we can show $(A + B)C = AC + BC$.

$\square$

(iv) Conclude that if R is a ring, then $M_n(R)$ is a ring.

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Given the content in the image, —

Given $(G, +)$ is an Abelian group,

$$\text{End}(G) = \{\varphi : G \to G \mid \varphi \text{ is a group homomorphism}\}$$

For $\varphi, \psi \in \text{End}(G)$,

$$(\varphi + \psi)(g) = \varphi(g) + \psi(g)$$

$$(\varphi \times \psi)(g) = \varphi(\psi(g))$$

(a)

Proof. We show the operations given above are well defined.

To show $\varphi + \psi \in \text{End}(G)$. Consider $g, h \in G$:

$$(\varphi + \psi)(g + h) = \varphi(g + h) + \psi(g + h) = \varphi(g) + \varphi(h) + \psi(g) + \psi(h)$$

Using the Abelian property:

$$= \varphi(g) + \psi(g) + \varphi(h) + \psi(h) = (\varphi + \psi)(g) + (\varphi + \psi)(h)$$

Thus, $\varphi + \psi \in \text{End}(G)$.

To show $\varphi \times \psi \in \text{End}(G)$. Consider $g, h \in G$:

$$(\varphi \times \psi)(g+h) = \varphi(\psi(g+h)) = \varphi(\psi(g)+\psi(h))$$

$$= \varphi(\psi(g)) + \varphi(\psi(h)) = (\varphi \times \psi)(g) + (\varphi \times \psi)(h)$$

Thus, $\varphi \times \psi \in \mathrm{End}(G)$. □

(b)

Theorem 0.1. Prove $(\mathrm{End}(G), +, \times)$ is a ring.

Proof. As G is an Abelian group, $(\mathrm{End}(G), +)$ is an Abelian group.

Associativity holds as composition of functions is associative.

Distributive property:

$$\varphi \times (\psi_1 + \psi_2)(g) = \varphi \times (\psi_1 + \psi_2)(g)$$

$$= \varphi(\psi_1(g) + \psi_2(g)) = \varphi(\psi_1(g)) + \varphi(\psi_2(g))$$

$$= (\varphi \times \psi_1)(g) + (\varphi \times \psi_2)(g)$$

$$= (\varphi \times \psi_1 + \varphi \times \psi_2)(g)$$

Thus, $(\mathrm{End}(G), +, \times)$ is a ring. □

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$$(\varphi_1 + \varphi_2) \times \varphi(g) = (\varphi_1 + \varphi_2)(\varphi(g)) = \varphi_1(\varphi(g)) + \varphi_2(\varphi(g)) = \varphi_1 \times \varphi(g) + \varphi_2 \times \varphi(g).$$

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Problem 3

Given: $(R, +, \cdot)$ is a ring; $a, b \in R$.

a) Show: $0 \cdot a = a \cdot 0 = 0$.

Proof. $0 \cdot a = (a - a) \cdot a = a^2 - a^2 = 0,$

$$a \cdot 0 = a \cdot (a - a) = a^2 - a^2 = 0. \quad \square$$

b) Show: $(na)b = n(ab) = a(nb)$.

Proof. Base case (n=1): Trivial.

$$\text{Induction Hypothesis: } (na)b = n(ab) = a(nb).$$

$$(a + \underbrace{a + \cdots + a}_{n+1 \text{ times}})b = (na + a)b = nab + ab = (n+1)ab,$$

$$a(\underbrace{b + \cdots + b}_{n+1 \text{ times}}) = a(nb + b) = nab + ab. \quad \square$$

c) If R is unital, prove: the multiplicative identity is unique.

Proof. Assume $1_A = 1_B$ are multiplicative identities. Then, $1_A = 1_A \cdot 1_B = 1_B$.

$$\text{Thus, the multiplicative identity is unique.}$$

$$\text{Show: } (-1) \cdot a = -a \text{ for any } a \in R.$$

$$\text{Proof: } a + (-1)a = 1 \cdot a + (-1)a = (1 + (-1))a = 0.$$

$$\Rightarrow (-1)a \text{ is the additive inverse } \Rightarrow (-1) \cdot a = -a. \quad \square$$

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Problem 4

Given: R is a commutative ring,

$$S, T \subseteq R,$$

$$ST = \bigcup_{n=1}^{\infty} \left\{ \sum_{i=1}^{n} s_i t_i \mid s_i \in S, t_i \in T \right\}.$$

a) If S and T are both subrings, show: ST is a subring.

Proof. Assume S and T to be non-empty. □

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Let $a, b \in ST$ such that $a = \sum_{i=1}^{n_1} s_i t_i$ and $b = \sum_{i=1}^{n_2} s'_i t'_i$.

Without loss of generality, assume $n_2 > n_1$. Then,

$$b - a = \sum_{i=1}^{n_2} s'_i t'_i - \sum_{i=1}^{n_1} s_i t_i = \sum_{i=n_1+1}^{n_2} s'_i t'_i + \sum_{i=1}^{n_1} (s'_i t'_i - s_i t_i).$$

By the Abelian property, we can rearrange the terms to get

$$\sum_{i=n_1+1}^{n_2} s'_i t'_i + \sum_{i=1}^{n_1} (s'_i t'_i - s_i t_i) = \sum_{i=1}^{n_2} s''_i t''_i.$$

Here, $i = 1$ to $i = n_1$ is set to 0 for the first entry, which implies

$$(ST, +) \prec (R, +).$$

Proof. Check the ring property:

$$\left(\sum_{i=1}^{n_1} s_i t_i \right) \left(\sum_{j=1}^{n_2} s'_j t'_j \right).$$

Without loss of generality, assume $n_1 < n_2$. Now set all terms to zero in the first entry after $i > n_1$,

$$\left(\sum_{i=1}^{n_2} s_i t_i \right) \left(\sum_{j=1}^{n_2} s'_j t'_j \right).$$

Every term will be of the form $s_i s'_j t_i t'_j$ for some $1 \leq i, j \leq n_2$. As R is a commutative ring, we get $s_i s'_j t_i t'_j$ and as S, T are subrings, $s_i s'_j \in S$ and $t_i t'_j \in T$ respectively. □

Definition 0.1. If S, T are both ideals, then ST is an ideal.

Proof. Consider $\sum_{i=1}^{n} s_i t_i \in ST$ and $r \in R$. Then,

$$r \left(\sum_{i=1}^{n} s_i t_i \right) = \sum_{i=1}^{n} (r s_i) t_i \quad \text{as } S \text{ is an ideal, we get } r s_i \in S \text{ and } \sum_{i=1}^{n} (r s_i) t_i \in ST.$$

Similarly,

$$\left(\sum_{i=1}^{n} s_i t_i \right) r = \sum_{i=1}^{n} s_i (t_i r) \quad \text{as } T \text{ is an ideal, we get } t_i r \in T \text{ and } \sum_{i=1}^{n} s_i (t_i r) \in ST.$$

□

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Given $m, n \in \mathbb{Z}$ and $d = \gcd(m, n)$.

Definition 0.2. Let I be the ideal generated by $\{m, n\}$.

Theorem 0.2. Show that $I = (d)$.

Proof. Consider $(X) = \bigcap_{J \in \mathcal{F}} J$, where

$$\mathcal{F} = \{J \triangleleft \mathbb{Z} \mid X \subseteq J\}.$$

If $a \in (X)$, then $a \in J$ for every $J \in \mathcal{F}$.

Since J is an ideal of $\mathbb{Z}$, it must also be a subgroup of $\mathbb{Z}$. Thus, $\exists n_J \in \mathbb{Z}$ such that $J = n_J\mathbb{Z}$.

If $X \subseteq n_J\mathbb{Z}$, then $n_J \mid x$ and $n_J \mid y$ for $x, y \in X$.

Since $d = \gcd(x, y)$, we have $d \mid n_J$.

Thus, $n_J\mathbb{Z} \supseteq d\mathbb{Z}$, where n_J is the integer corresponding to J.

Therefore, $(X) \supseteq d\mathbb{Z}$.

$d\mathbb{Z}$ is the smallest ideal containing X.

Consider $n\mathbb{Z} \triangleleft \mathbb{Z}$ such that $X \subseteq n\mathbb{Z}$ and $n\mathbb{Z} \subseteq d\mathbb{Z}$. Let $nk_1 = x$ and $nk_2 = y$.

By Bézout's identity,

$$nk_1r_1 + nk_2r_2 = xr_1 + yr_2 = d,$$

since $d = \gcd(x, y)$ and $k_1, k_2, r_1, r_2 \in \mathbb{Z}$.

Thus, $n \mid d$, which implies $d\mathbb{Z} \subseteq n\mathbb{Z}$.

Hence, $d\mathbb{Z} = n\mathbb{Z}$. □

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Given the content in the image, —

Definition 0.3. Let $C_c(\mathbb{R})$ be the set of continuous functions $f : \mathbb{R} \to \mathbb{C}$ with compact support, i.e.,

$$C_c(\mathbb{R}) := \{f : \mathbb{R} \to \mathbb{C} \mid f \text{ is continuous and } \exists a, b \in \mathbb{R} \text{ such that } f(x) = 0 \text{ whenever } x \notin [a, b]\}.$$

Here, $+$ and $\times$ are defined as usual addition and multiplication of functions.

Remark 0.1. Observe that $C_c(\mathbb{R})$ is a ring and is non-empty.

Theorem 0.3. Show that $C_c(\mathbb{R})$ is not unital.

Proof. Assume there exists $g \in C_c(\mathbb{R})$ such that $f \times g = f$ for any $f \in C_c(\mathbb{R})$.

If $g(x) \neq 0$ for some $x \in [a, b]$, choose $f \in C_c(\mathbb{R})$ such that $f(x) \neq 0$ for some $x \in [a', b']$ where $[a, b] \subset [a', b']$.

Consider $x_0 = \frac{a'+a}{2}$. Then,

$$g(x_0) = 0 \quad \text{and} \quad f(x_0) \neq 0 \implies f \times g(x_0) \neq 0.$$

This leads to a contradiction since $f \times g = f$ implies $f \times g(x_0) = f(x_0) \neq 0$ but $g(x_0) = 0$ implies $f \times g(x_0) = 0$. Hence, no unit exists in $C_c(\mathbb{R})$. $\square$

Given the handwritten content, —

1. Given: R is a ring, S be a subring of R and $I \trianglelefteq R$.

   (a) *Proof.* Prove: $S + I := \{s + a \mid s \in S, a \in I\}$ is a subring.

   Proof. Observe, $S + I$ is non-empty. Then consider $a', b' \in S + I$ where $a' = s_1 + a$ and $b' = s_2 + b$. Now,

   $$a' - b' = s_1 + a - s_2 - b = (s_1 - s_2) + (a - b) \in S + I$$

   where $s_1 - s_2 \in S$ and $a - b \in I$.

   Therefore, $(S + I, +) \leq (R, +)$.

   Now, let $a', b' \in S + I$ where $a' = s_1 + a$, $b' = s_2 + b$,

   $$a' \cdot b' = (s_1 + a)(s_2 + b) = s_1 s_2 + a s_2 + s_1 b + ab$$

   where $s_1 s_2 \in S$ and $a s_2 + s_1 b + ab \in I$.

   Thus, $(S + I, +, \cdot)$ is a subring of $(R, +, \cdot)$. $\square$

   (b)

   Theorem 0.4. Prove: $I \trianglelefteq S + I$.

   Proof. Consider $a \in I$ and $b' \in S + I$ such that $b' = s + b$ where $s \in S$, $b \in I$. Observe

   $$ab' = a(s + b) = as + ab,$$

   as $I \trianglelefteq R$, $as \in I$ and $ab \in I \Rightarrow ab' \in I$.

   Similarly, $b'a \in I$. $\square$

   (c)

   Theorem 0.5. Show: $S \cap I \trianglelefteq S$.

   Proof. As $(I, +) \trianglelefteq R$,

   $$(S \cap I, +) \trianglelefteq S \quad \text{and} \quad 0 \in S \cap I \Rightarrow S \cap I \neq \emptyset.$$

   If $a \in S \cap I$ and $s \in S$, then $a \in I$ and $s \in S \Rightarrow as \in I$.

   Now, $a \in S$ and $s \in S \Rightarrow as \in S$. $\square$

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$$\text{Hence, as } s \in S \cap I \implies \varphi(s) = 0 + I \implies S \cap I \subseteq \ker\varphi.$$

$$\text{If } s \in \ker\varphi, \varphi(s) = I \implies s \in S \text{ and } s \in I \implies s \in S \cap I.$$

$$\therefore S \cap I = \ker\varphi.$$

Proof. (cd) Prove: $\frac{S+I}{I} \cong \frac{S}{S \cap I}$

(1) Well definedness

$$s_1 = s_2 \implies s_1 + I = s_2 + I$$

(2) φ is a ring homomorphism

$$\varphi(s + s') = s + s' + I = (s + I) + (s' + I) = \varphi(s) + \varphi(s')$$

$$\varphi(ss') = ss' + I = (s + I)(s' + I) = \varphi(s)\varphi(s')$$

(3) φ is surjective (By construction)

(4) $\ker\varphi = S \cap I$

$$\text{If } s \in S \cap I, \text{ now } \varphi(s) = 0 + I \text{ as } s \in S \cap I$$

Hence, $S \cap I \subseteq \ker\varphi$.

If $s \in \ker\varphi$, $\varphi(s) = I$, now $s \in S$ and $s \in I \implies s \in S \cap I$.

$$\therefore S \cap I = \ker\varphi.$$

By First Isomorphism Theorem,
$$\frac{S + I}{I} \cong \frac{S}{S \cap I}.$$

□

Given the content in the image, —

1. Given: $D \in \mathbb{Q}$ be a rational number and is NOT a perfect square.

$$R := \{a + b\sqrt{D} \mid a, b \in \mathbb{Q}\}$$

(a) *Proof.* Show: R is a subring of $\mathbb{C}$.

Observe, $R \neq \emptyset$ as $0 \in R$.

Consider $x, y \in R$, where $x = a_1 + b_1\sqrt{D}$ and $y = a_2 + b_2\sqrt{D}$; $a_1, b_1, a_2, b_2 \in \mathbb{Q}$.

$$x - y = (a_1 - a_2) + (b_1 - b_2)\sqrt{D} \in R \quad \text{as} \quad a_1 - a_2 \in \mathbb{Q} \quad \text{and} \quad b_1 - b_2 \in \mathbb{Q}.$$

Thus, $(R, +)$ is a subgroup of $(\mathbb{C}, +)$.

Now, look at

$$xy = (a_1 + b_1\sqrt{D})(a_2 + b_2\sqrt{D}) = a_1a_2 + b_1b_2D + (a_2b_1 + a_1b_2)\sqrt{D}.$$

Since $a_1a_2 + b_1b_2D \in \mathbb{Q}$ and $a_2b_1 + a_1b_2 \in \mathbb{Q}$, it follows that $xy \in R$.

Hence, $(R, +, \cdot)$ is a subring of $(\mathbb{C}, +, \cdot)$. $\square$

(b)

Theorem 0.6. If $(a + b\sqrt{D}) \neq 0$, then show that $a^2 - b^2D \neq 0$.

Proof. If $(a + b\sqrt{D}) \neq 0$, then $a \neq 0$ or $b \neq 0$.

Hence, $a - b\sqrt{D} \neq 0$. Now,

$$(a + b\sqrt{D})(a - b\sqrt{D}) \neq 0 \implies a^2 - b^2D \neq 0.$$

$\square$

(c)

Theorem 0.7. Prove: R is a field.

Proof. Observe R is a commutative, unital ring.

Moreover, R is a division ring as for $a + b\sqrt{D} \neq 0$, then

$$\frac{a - b\sqrt{D}}{a^2 - b^2D}$$

is the inverse with respect to multiplication.

Now, R is a commutative division ring, which is a field. $\square$

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Given R is a unital ring,

$$Z(CR) := \{a \in R \mid ab = ba \text{ for all } b \in R\}.$$

Proof. Observe, $Z(CR) \neq \emptyset$ as $1 \in Z(CR)$.

Now, consider $a, a' \in Z(CR)$.

For any $b \in R$, $(a - a')b = ab - ab' = ba - ba' = b(a - a') \implies (Z(CR), +)$ is a subgroup of R.

For any $b \in R$, observe $aa'b = aba' = baa' \implies aa' \in Z(CR)$.

Hence, $Z(CR)$ is a subring.

$\square$